REPUBLIC OF THE PHILIPPINES

DEPARTMENT OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

# PHILIPPINE STARTUP CHALLENGE XI

TEAM NAME: **SIPAT LABS**

STARTUP NAME: **SeePat**

## CONCEPT NOTE

## I. SUMMARY

**SeePat** is derived from **SIPAT - Smart Insights for Profit, Assets and Trade**. The name draws from the Filipino word ***sipat***, meaning to look closely, observe, check, or assess carefully. This reflects SeePat's purpose: helping microentrepreneurs look beyond daily sales and understand what is actually happening in their businesses.

SeePat is a proposed voice-first financial and business decision-intelligence platform designed for Filipino microentrepreneurs, including sari-sari store owners, carinderias and food stalls, public-market vendors, home-based sellers, neighborhood retailers, and other owner-managed microbusinesses.

The startup addresses a practical problem: knowing how much was sold does not necessarily mean knowing how much was earned, where working capital went, how much is tied up in customer credit, or what should be restocked. Sales, expenses, inventory, utang, and capital are often recorded separately or remembered informally.

*"Magkano talaga ang kinita ko?"*

*"Nasaan ang puhunan ko?"*

*"Bakit mataas ang benta pero kulang ang cash?"*

*"Magkano ang nasa utang?"*

*"Ano ang mabilis at mabagal mabenta?"*

*"Ano ang dapat kong i-restock?"*

SeePat proposes a Voice-to-Business interaction model that converts natural Filipino or English business speech into validated, structured transaction records. For example, an owner may say, "Nakabenta ako ng sampung Softdrinks, tig-bente." The proposed system will interpret the statement, ask for confirmation when needed, record the transaction, update related business records, and feed verified data into deterministic financial computations.

SeePat is not positioned as another Point-of-Sale system. Its primary value is helping entrepreneurs understand what their business numbers mean and use those numbers to

make better decisions: POS records what happened; bookkeeping calculates what happened; SeePat is designed to help explain why it happened, where the money went, and what the entrepreneur should consider doing next.

Current Development Stage: Prototyping. SIPAT LABS has defined the problem, target users, value proposition, business model, core workflows, product architecture, feature scope, and prototype design. A functional Minimum Viable Product (MVP) has not yet been completed. Problem validation, prototype validation, functional MVP development, controlled testing, and pilot deployment are planned as succeeding milestones.

## II. BACKGROUND OF THE PROBLEM

Micro, small, and medium enterprises (MSMEs) constitute a vital pillar of the Philippine economy, contributing significantly to entrepreneurship, employment, and local economic activity. According to government-reported 2024 MSME statistics, MSMEs comprise 1,236,908 establishments, representing 99.63% of all business establishments in the Philippines. Microenterprises alone account for 1,125,476 establishments, or 90.66% of the total, underscoring their dominant presence in the country's business landscape. These figures highlight the critical role of microenterprises and the importance of strengthening their capacity, productivity, resilience, and long-term sustainability. *(Source: Philippine Information Agency (PIA), citing DTI/PSA 2024 MSME statistics)*

Despite their economic importance, many microentrepreneurs continue to operate with limited financial-management and digital decision-support capabilities. DTI's MSME digitalization research has also identified gaps in the adoption of advanced digital tools and substantial capacity-building requirements among MSMEs, including financial-management capabilities. *(Department of Trade and Industry. Baseline Survey on Digitalization of MSMEs in the Philippines)*

For many small neighborhood businesses, the challenge is not simply the absence of technology. The deeper problem is the disconnect between everyday transactions and actionable business understanding.

Traditional notebooks, calculators, memory-based recording, and fragmented digital records make these questions difficult to answer consistently.

Existing digital POS and bookkeeping applications have significantly improved transaction recording, inventory management, and financial reporting. However, this creates another opportunity: helping entrepreneurs interpret their business information without requiring them to become accountants, analysts, or expert users of complex dashboards.

Government programs promoting sari-sari store and MSME digitalization further demonstrate the relevance of developing accessible digital solutions for this sector. *(Department of Trade and Industry. Tindahan Mo, e-Level Up Mo!)*

The opportunity is therefore not merely to digitize transactions, but to make business data understandable and actionable for owners who may not have accountants, analysts, or sophisticated business systems.

## III. PROPOSED STARTUP SOLUTION

SeePat is a mobile-first, voice-first financial and business decision-intelligence platform designed around a simple operating cycle:

**SPEAK/RECORD → VALIDATE → COMPUTE → UNDERSTAND → PREDICT → DECIDE**

The system combines structured business recording, deterministic financial computations, natural-language processing, speech recognition, analytics, forecasting, and explainable AI.

### A. Voice-to-Business

A key proposed innovation of SeePat is Voice-to-Business.

Rather than treating speech recognition merely as voice-to-text, SeePat is designed to transform natural business speech into structured and actionable business information.

The proposed process is:

**Speech → Transcription → Intent Recognition → Entity Extraction → User Validation → Structured Transaction → Financial Computation → Business Record → Insight**

For example:

*"May utang si Aling Nena—dalawang sardinas at isang kilong asukal."*

SeePat may identify:

Customer: Aling Nena

Transaction: Credit Sale

Items: 2 sardines + 1 kg sugar

Payment Status: Receivable / Utang

After user confirmation, the transaction can automatically update sales, inventory, customer receivables, and the business's financial position.

Voice interaction is intended to reduce the friction associated with conventional bookkeeping interfaces, particularly for entrepreneurs who are accustomed to managing their businesses conversationally and informally.

Manual forms remain available for users who prefer conventional input or when voice recognition is uncertain.

**B. SeePat Ledger**

Records and organizes sales, purchases, operating expenses, cash inflows, cash outflows, and transaction history.

**C. SeePat Stock**

Provides inventory visibility into current stock, inventory movement, fast and slow-moving products, low-stock indicators, estimated stock depletion and proposed replenishment guidance.

**D. SeePat Utang**

Tracks customer receivables, outstanding balances, payments, aging of utang and total capital currently tied up in receivables.

**E. SeePat Capital — "Nasaan ang Puhunan Ko?"**

This is one of SeePat's signature features. Help the owner understand how working capital is distributed across cash, inventory, receivables, and relevant liabilities rather than showing cash alone.

This helps answer:

> *"Nasaan ang puhunan ko?"*

For example, SeePat may explain:

> *"Your available cash decreased this week because ₱3,200 remains in customer receivables while ₱5,600 was used to replenish inventory. A significant portion of your working capital is therefore currently held in inventory and utang rather than cash."*

**F. SeePat Insights**

Converts verified records into understandable explanations of sales, expenses, profit, inventory, receivables, cash position, and capital allocation.

Instead of simply presenting:

> Sales: ₱8,420

Expenses: ₱5,730

the platform may explain:

> *"Sales increased this week, but estimated profit did not increase at the same rate because product acquisition and operating expenses also increased."*

The purpose is to move from data display to business understanding.

**G. SeePat Score**

A proposed explainable business-health indicator combining selected operational and financial signals such as profitability, cash position, receivable exposure, inventory movement, expense behavior and sales trends.

The score will not be presented as a substitute for professional financial assessment. Its purpose is to provide entrepreneurs with a simplified indicator of areas requiring attention, together with an explanation of the factors influencing the score.

**H. Ask SeePat**

Users can ask natural-language questions such as:

> *"Magkano kinita ko ngayong linggo?"*
>
> *"Bakit bumaba ang tubo ko?"*
>
> *"Magkano ang nasa utang?"*
>
> *"Ano ang pinakamabilis mabenta?"*
>
> *"Ano ang matagal nang hindi gumagalaw?"*
>
> *"Nasaan ang puhunan ko?"*
>
> *"Ano ang dapat kong i-restock?"*

The system retrieves the relevant verified business records, performs the required calculations, and generates a simplified explanation.

**I. Explainable Decision Intelligence**

Recommendations will be accompanied by evidence and business factors behind them.

Instead of:

> *"Restock Softdrinks."*

SeePat will provide an explanation such as:

> *"Softdrinks averaged 18 units sold per day during the last seven selling days. You currently have 12 units remaining. At the recent sales rate, your inventory may be depleted within one selling day."*

This approach allows the entrepreneur to understand why a recommendation was generated.

**J. Business Twin**

Each verified transaction will update a simplified digital representation of the enterprise covering sales, expenses, cash, inventory, receivables, payables, and profitability. This Business Twin becomes the foundation for financial calculations, conversational queries, business-health indicators, forecasts, and recommendations.

**K. Localized Financial Experience**

SeePat will initially prioritize Filipino and English business language, particularly familiar terms such as:

> *benta, kita, tubo, puhunan, utang, gastos, paninda, bayad, and sukli.*

Future versions may expand support for regional Philippine languages based on validated market demand.

The architecture deliberately separates calculation from AI: validated formulas compute financial values, while AI performs language understanding, pattern detection, forecasting, explanation, and recommendation. This reduces hallucination risk and makes advice traceable to actual business records.

**SDG Alignment:**

***SDG 8 – Decent Work and Economic Growth***, particularly entrepreneurship, productive activities, and MSME development

***SDG 9 – Industry, Innovation and Infrastructure*** through accessible digital transformation

***SDG 10 – Reduced Inequalities*** by making business intelligence accessible to entrepreneurs who cannot afford professional accounting and analytics systems

## IV. OBJECTIVES

### General Objective

To develop and validate SeePat as an accessible voice-first financial and business decision-intelligence platform that helps Filipino microentrepreneurs transform everyday business transactions into understandable and actionable business information.

### Specific Objectives

1. Validate the identified business problems, current record-keeping practices, and decision-support needs of target microentrepreneurs before functional MVP development.
2. Develop and evaluate the proposed Voice-to-Business workflow for transforming natural-language business statements into user-validated structured transactions.
3. Develop a functional mobile-first MVP after initial problem and prototype validation, integrating sales, expenses, inventory, receivables, financial computation, capital visibility, voice-based transaction capture, and explainable business insights.
4. Validate deterministic financial computations against manually verified test cases and target at least 95% computational accuracy before pilot deployment.
5. Target at least 80% unaided completion of identified essential workflows during usability validation.
6. Conduct an initial pilot involving approximately 30 target microbusinesses, subject to participant availability and validation requirements.
7. Measure whether SeePat improves users' ability to identify profit, inventory position, outstanding receivables, and capital allocation compared with their existing practices.
8. Evaluate product usability, perceived usefulness, continued-use intention, retention, and willingness to pay.
9. Develop a scalable, privacy-aware, secure, and offline-tolerant architecture suitable for subsequent commercialization and institutional deployment..

## V. TARGET MARKET/BENEFICIARIES

SeePat's primary target users are Filipino microentrepreneurs whose operations involve frequent small transactions and who may not employ dedicated accountants or use sophisticated business-management systems.

Primary beneficiaries include:

- sari-sari store owners;
- carinderias;
- food stalls;
- public-market vendors;
- home-based sellers;
- small neighborhood retailers;
- micro service businesses; and
- other owner-managed microenterprises.

Secondary institutional customers and partners may include:

- cooperatives;
- local government units;
- livelihood programs;
- microfinance institutions;
- entrepreneurship organizations;
- business incubators;
- MSME development organizations; and
- government agencies implementing livelihood and digitalization initiatives.

The initial market-entry strategy will prioritize a geographically concentrated validation and pilot area so the team can closely observe user behavior, test assumptions, refine the product, and gather evidence before broader expansion.

## VI. VALUE PROPOSITION

The Philippine market already contains capable POS, bookkeeping, inventory, and MSME digitalization solutions.

For example, Peddlr provides functionality covering POS, inventory, expenses, ledgers, reports, and offline operation. Packworks provides digital tools for sari-sari stores and has participated in DOST-supported work involving machine-learning-based customized recommendations.

These existing solutions demonstrate both market demand and the importance of avoiding an unsupported claim that SeePat is the first platform to combine digital business management and AI.

SeePat is therefore not positioned as another POS or bookkeeping application.

Its proposed differentiation is:

**Localized, voice-first, explainable business decision intelligence for microentrepreneurs.**

**Competitive Positioning**

| Existing Alternative | Existing Strength | SeePat's Proposed Differentiation |
|---|---|---|
| **Peddlr** | POS, inventory, expenses, ledgers, reports and offline operation | Voice-to-Business, conversational diagnosis, Business Twin, explainable insights and capital intelligence |
| **Packworks** | Sari-sari digitization, inventory/sales capabilities and emerging AI/ML recommendation capabilities | Owner-facing natural-language financial intelligence, voice transaction capture, explainable reasoning and capital decomposition |
| **Bookkeeping applications** | Recording income, expenses, receivables and financial records | Moves from recording and calculation toward conversational interpretation and decision support |
| **Full POS platforms** | Comprehensive retail operations, inventory and analytics | Lightweight decision intelligence designed around the needs of owner-managed microenterprises |
| **Notebook / Calculator** | Familiar, inexpensive and flexible | Automatically connects transactions to profit, inventory, receivables, capital and business insights |
| **General-purpose AI chatbot** | Strong conversational capability and general business advice | Answers are grounded in the entrepreneur's own validated business records and deterministic financial computations |

**Five Core Differentiation Pillars**

**1. Voice-to-Business**

Natural Filipino or English business speech is transformed into user-validated structured business records, not merely transcribed into text.

**2. Conversational Business Intelligence**

Entrepreneurs can ask ordinary questions about their negosyo rather than having to interpret complex dashboards.

**3. Explainable Decision Intelligence**

SeePat shows the evidence and business factors underlying recommendations.

**4. Business Twin and Capital Intelligence**

Transactions continuously update the enterprise's financial and operational representation, enabling the system to answer questions such as “Nasaan ang puhunan ko?”

**5. Hyperlocal Financial UX**

The experience is designed around familiar Filipino microbusiness vocabulary and operating realities rather than simply translating a conventional enterprise interface.

The differentiation can be summarized as: POS records what happened. Bookkeeping calculates what happened. SeePat is designed to help explain why it happened, where the money went, and what the entrepreneur should consider doing next.

## VII. BUSINESS MODEL

SeePat proposes a **freemium SaaS** model intended to reduce adoption barriers while creating recurring revenue from higher-value capabilities. Pricing remains proposed and will be tested through market validation and willingness-to-pay studies.

**SeePat FREE - ₱0/month**

- Core transaction recording
- Basic sales and expense tracking
- Basic inventory
- Utang tracking
- Daily business summary

**SeePat PRO - Proposed ₱99/month**

- AI business insights
- Advanced profit analysis
- SeePat Score
- Voice-to-transaction
- Demand/reorder recommendations
- Cloud backup
- Advanced reports

**SeePat NEGOSYO — Proposed ₱199/month**

- Multiple users
- Advanced analytics

- Supplier analysis
- Enhanced forecasting
- Multi-device access
- Expanded reports

**Institutional packages** may provide bulk licenses and management dashboards for cooperatives, LGUs, livelihood programs, and partner organizations. Potential future revenue sources include SaaS subscriptions, institutional licensing, premium analytics, and partner integrations

## VIII. MARKET ANALYSIS

The addressable market is large and economically important: microenterprises dominate Philippine establishments *(Source: Philippine Information Agency (PIA), citing DTI/PSA 2024 MSME statistics),* while national programs continue to promote MSME digitalization *(Sources: Department of Trade and Industry (DTI), Baseline Survey on Digitalization of MSMEs in the Philippines, September 2020 and Department of Trade and Industry (DTI), "Tindahan Mo, e-Level Up Mo!")*. The market is competitive, which validates demand but requires precise positioning. Peddlr states that it has helped more than two million Filipino stores, while DOST-PCIEERD reported Packworks' network at over 270,000 sari-sari stores and documented its AI/ML initiative for customized recommendations. SeePat will therefore avoid competing on basic POS features. Its proposed beachhead is microentrepreneurs who need **low-friction**, **explainable business decision support** and prefer **conversational interaction over complex accounting dashboards**.

The initial strategy is to **validate locally - establish product-market fit - partner with institutions - replicate regionally - scale nationally**. Competitive advantage will be evaluated through evidence such as whether users can more quickly and correctly answer core business questions, complete essential workflows, understand recommendations, continue using the product, and demonstrate willingness to pay.

## IX. OPERATIONS PLAN

SeePat will follow a **Lean Startup and evidence-driven product-development** approach.

### Phase 1 - Problem and Prototype Validation

1. Conduct structured interviews with target microentrepreneurs.
2. Document current practices for sales, expenses, inventory, utang, profit determination, and purchasing decisions.

3. Validate whether the identified pain points, terminology, proposed workflows, and core value proposition reflect actual user needs.
4. Use prototype demonstrations or mockups to collect early usability and comprehension feedback before building the functional MVP.

**Phase 2 – Functional MVP Development**

After initial validation, develop a mobile-first PWA using a scalable web technology stack. The planned MVP will initially include Authentication, Dashboard, Sales, Expenses, Inventory, Utang (Credits), Profit Engine, SeePat Score, Voice-to-Business transaction capture, and AI Insights.

**Phase 3 - Controlled Testing**

1. Validate financial computations against manually calculated test cases.
2. Constrain AI-generated explanations and recommendations to verified business data.
3. Evaluate Voice-to-Business interpretation and user-confirmation workflows.
4. Measure task completion, errors, user comprehension, and satisfaction.

**Phase 4 - Pilot Deployment**

1. Deploy the MVP to approximately 30 participating microbusinesses, subject to availability and validation requirements.
2. Collect usage metrics, transaction data with consent, usability feedback, feature requests, retention indicators, and willingness-to-pay evidence.

**Phase 5 - Product Refinement**

Improve the application based on pilot evidence.

Prioritize features that demonstrate measurable user value rather than adding unnecessary functionality.

**Phase 6 - Commercialization**

Following successful validation and refinement, introduce the proposed freemium subscription model and pursue partnerships with cooperatives, MSME organizations, livelihood programs, and other institutional partners.

**Data Protection**

SeePat will incorporate privacy-by-design principles, role-based access controls, secure authentication, encryption where appropriate, data minimization, consent mechanisms, auditability, and compliance with applicable Philippine data-protection requirements.

## X. FINANCIAL REQUIREMENT

An initial funding requirement of ₱150,000 is proposed to build, validate, and pilot SeePat after the current prototyping stage.

| **Use of Funds** | **Amount** | **Purpose** |
|---|---|---|
| Functional MVP Development & Technical Infrastructure | ₱50,000 | Mobile-first PWA, database/cloud services, Voice-to-Business processing, financial computation engine, AI integration, security, domain and deployment |
| User Research & Validation | ₱15,000 | Field interviews, workflow validation, usability testing, transportation, documentation and user feedback |
| Pilot Deployment & Onboarding | ₱20,000 | Deployment to initial microbusiness users, onboarding, training materials and pilot support |
| AI, Speech & Cloud Services | ₱40,000 | Speech-to-text, LLM/API usage, database, storage, monitoring and other consumption-based services during pilot |
| Testing, QA & Data Privacy | ₱10,000 | Functional testing, financial-computation validation, security checks and privacy-compliance preparation |
| Early Commercialization & Marketing | ₱10,000 | Branding, promotional materials, demonstrations, merchant acquisition and initial partnership activities |
| Contingency | ₱5,000 | Unanticipated validation, pilot and technical expenses |
| **TOTAL** | **₱150,000** | |